

GRAVEATLAS

Phase 16 — AI-Native Experience & Intelligent GUI

GRAVEATLAS — PHASE 16
AI-NATIVE EXPERIENCE & INTELLIGENT GUI

PHASE 1 THROUGH PHASE 15 MUST ALREADY BE COMPLETE.

OBJECTIVE

Redesign the GraveAtlas user experience for the AI era.
The GUI must feel like an intelligent historical research environment,
not a conventional database administration interface.

CORE RULES

- Do NOT rebuild Phases 1-15.
- Do NOT start Phase 17.
- Reuse existing backend, data, provenance, security, preservation, automation and intelligence systems.
- No paid AI/cloud services or external integrations without explicit approval.
- Prefer existing/free/local capabilities where practical.
- Never expose secrets, credentials, tokens or private data.
- Never fabricate historical facts, records, sources, institutions, analytics, performance or AI results.
- AI inference is never automatically authoritative historical truth.
- Do not perform destructive actions without authorization.

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PART 1 — PHASE 15 BASELINE
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Read the actual Phase 15 implementation and final report.

Identify:

- existing GUI
- navigation
- search
- map
- records
- AI capabilities
- provenance UI
- mobile limitations
- accessibility limitations
- performance limitations
- technical debt

Do not redesign blindly.

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PART 2 — AI COMMAND CENTER
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Create a primary natural-language interface.

Examples:

"Find cemeteries in Singapore established before 1900."
"Help me investigate this cemetery."
"Show the sources supporting this record."
"Compare these historical records."
"Find gaps in the evidence."

Support where available:

- text input
- voice input
- suggested prompts
- conversation history
- contextual follow-up
- current-screen awareness

AI responses must remain evidence-linked.

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PART 3 — CONVERSATIONAL RESEARCH WORKSPACE
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Design the flow:

ASK → EXPLORE → EVIDENCE → COMPARE → SAVE

Users must be able to move from AI questions to actual records,
sources, maps and research collections.

Do not trap users inside chat.

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PART 4 – EVIDENCE-FIRST RESULTS

Clearly distinguish:

KNOWN
SOURCE-BACKED
INFERRED
UNCERTAIN
CONFLICTING
NEEDS VERIFICATION

Provide direct access to supporting evidence.

Never present inference as fact.

PART 5 – AI COPILOT

Provide contextual assistance for:

- searching
- explaining records
- comparing sources
- summarizing evidence
- identifying gaps
- suggesting research questions
- navigating GraveAtlas

Respect permissions and privacy.

PART 6 – AI MAP

Make the map an intelligent research surface.

Where supported:

- natural-language map queries
- relevant-location highlighting
- cemetery clusters
- historical layers
- timeline controls
- source overlays
- "What changed here?"
- "Explain this area."

Geographic conclusions must remain source-aware.

PART 7 – RESEARCH CANVAS

Create a visual workspace for:

PERSON → CEMETERY → RECORD → SOURCE → PLACE → INSTITUTION → EVENT

Allow:

- connecting records
- evidence inspection
- notes
- saved findings
- source comparison
- accepting/rejecting AI suggestions

AI relationships remain non-authoritative until reviewed.

PART 8 – MULTIMODAL RESEARCH

Where supported, allow:

- text
- photographs
- scanned documents
- maps
- OCR
- voice questions

Clearly separate:

OBSERVED
FROM SOURCE
INFERRED
UNKNOWN

Never fabricate unreadable text.

PART 9 – SMART RECORD CARDS

Use clear, expandable cards containing:

- name
- evidence status
- location
- date
- sources
- related records
- provenance
- actions

Do not hide important provenance.

PART 10 – AI TIMELINES

Create interactive timelines where supported.

Each important timeline item should connect to:

DATE → EVENT → RECORD → SOURCE

Do not generate unsupported events.

PART 11 – ADAPTIVE INTERFACE

Support task-oriented modes:

RESEARCH MODE

AI + evidence + timeline + graph

MAP MODE

map + geographic AI + historical layers

ARCHIVE MODE

datasets + provenance + preservation

INSTITUTION MODE

collections + verification + governance

PUBLIC MODE

simple search + maps + cemetery exploration

Avoid duplicate or confusing interfaces.

PART 12 – MOBILE-FIRST GUI

Optimize for phones:

- persistent AI command bar
- voice button where supported
- swipeable cards
- map/research toggle
- minimal navigation
- large touch targets
- responsive layouts
- accessible controls

Do not sacrifice evidence for visual simplicity.

PART 13 – DESKTOP GUI

Provide efficient research layouts:

- map + evidence
- graph + record
- AI + sources
- comparison views
- split-screen research

Avoid excessive panels.

PART 14 – AI TRANSPARENCY

Provide "WHY AM I SEEING THIS?" for important AI results.

Show where appropriate:

- sources used

- records used
- evidence category
- uncertainty
- AI workflow/model information where available
- timestamp

Never expose private credentials.

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PART 15 – CONTEXTUAL ACTIONS
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For records provide relevant actions:

- Explain this record
- Show sources
- Find related records
- Compare
- Add to research
- View on map
- View timeline
- Check provenance

Respect permissions.

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PART 16 – VOICE
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Where supported:

- voice questions
- spoken navigation
- spoken research queries

Require clear confirmation before important actions.
Never execute destructive actions from ambiguous voice input.

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PART 17 – ACCESSIBILITY
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Review:

- screen readers
- keyboard navigation
- contrast
- text scaling
- touch targets
- reduced motion
- captions/transcripts
- voice alternatives

Use an appropriate accessibility standard where practical.

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PART 18 – AI ERROR HANDLING
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When uncertain:

- say so
- show evidence
- ask for clarification
- offer alternative searches

Never fill missing evidence with invented content.

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PART 19 – PERFORMANCE
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Measure where practical:

- initial load
- AI response rendering
- map rendering
- graph rendering
- search
- image/document processing
- mobile performance

Use actual measurements only.

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PART 20 – PRIVACY
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Review:

- conversation history
- research history

- uploaded documents
- voice data
- saved searches
- user behavior
- institutional data

Apply data minimization and appropriate retention.

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PART 21 – SECURITY

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Review:

- AI input handling
- prompt injection
- tool permissions
- contextual data access
- uploaded files
- voice input
- conversation storage
- API access

AI must not bypass existing authorization.

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PART 22 – DESIGN SYSTEM

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Create/update a consistent design system covering:

- typography
- spacing
- cards
- buttons
- navigation
- evidence badges
- confidence indicators
- maps
- timelines
- graph elements
- dialogs
- loading states
- error states

Keep the visual language modern, calm and research-oriented.

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PART 23 – AI-NATIVE HOME SCREEN

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The home screen should ask:

"What would you like to investigate?"

Include:

- AI command input
- voice option
- suggested research tasks
- recent research
- saved investigations
- relevant discovery where permission allows

Do not overwhelm first-time users.

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PART 24 – RESEARCH MEMORY

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Where privacy permits, allow continuation of research sessions.

Remember:

- current investigation
- selected records
- saved evidence
- research notes
- previous questions

Users must be able to inspect and delete research history.

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PART 25 – UI TESTING

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Test:

- first-time user flow
- experienced researcher flow
- mobile flow

- desktop flow
- accessibility
- AI uncertainty
- evidence inspection
- map interaction
- record exploration
- error recovery

Use actual results.

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PART 26 — DOCUMENTATION

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Create/update:

docs/AI-NATIVE-GUI.md
 docs/AI-COMMAND-CENTER.md
 docs/CONVERSATIONAL-RESEARCH.md
 docs/EVIDENCE-FIRST-UI.md
 docs/AI-COPILOT.md
 docs/AI-MAP.md
 docs/RESEARCH-CANVAS.md
 docs/MULTIMODAL-RESEARCH.md
 docs/SMART-RECORD-CARDS.md
 docs/AI-TIMELINES.md
 docs/ADAPTIVE-INTERFACE.md
 docs/MOBILE-GUI.md
 docs/DESKTOP-GUI.md
 docs/AI-TRANSPARENCY.md
 docs/CONTEXTUAL-ACTIONS.md
 docs/VOICE-EXPERIENCE.md
 docs/ACCESSIBILITY.md
 docs/AI-ERROR-HANDLING.md
 docs/DESIGN-SYSTEM.md
 docs/RESEARCH-MEMORY.md
 docs/PHASE-16-ROADMAP.md

Document actual implementation only.

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PHASE 16 ACCEPTANCE GATE

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[] Phase 15 baseline reviewed
 [] AI command center
 [] Conversational research
 [] Evidence-first results
 [] AI copilot
 [] AI map
 [] Research canvas
 [] Multimodal research
 [] Smart record cards
 [] AI timelines
 [] Adaptive interface
 [] Mobile-first GUI
 [] Desktop GUI
 [] AI transparency
 [] Contextual actions
 [] Voice experience
 [] Accessibility
 [] AI error handling
 [] Performance evaluation
 [] Privacy review
 [] Security review
 [] Design system
 [] AI-native home screen
 [] Research memory
 [] UI testing
 [] Documentation

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FINAL REPORT

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PHASE 16 STATUS: READY / NOT READY

Report:

- what was built
- evidence
- tests/screenshots where available
- performance measurements
- accessibility results
- security findings
- privacy findings
- limitations

- unresolved issues

Use actual results only.

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UX SCORECARD

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AI usability:
[actual assessment]

Evidence clarity:
[actual assessment]

Research efficiency:
[actual assessment]

Mobile usability:
[actual assessment]

Desktop usability:
[actual assessment]

Accessibility:
[actual assessment]

AI transparency:
[actual assessment]

Performance:
[actual assessment]

Security:
[actual assessment]

Privacy:
[actual assessment]

Do not invent numeric scores without a defined methodology.

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BLOCKERS

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For each blocker report:

- severity
- component
- issue
- evidence
- impact
- remediation
- status

Never include secrets.

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ROADMAP

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NOW:
[items]

NEXT:
[items]

LATER:
[items]

DO NOT BUILD YET:
[items]

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STOP CONDITION

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Do NOT start Phase 17.

Do NOT add paid AI/cloud services without explicit approval.

Do NOT expose secrets.

Do NOT fabricate historical information.

Do NOT allow unsupported AI claims to become authoritative.

Do NOT bypass authorization through AI tools.

Do NOT perform destructive actions from ambiguous natural-language or voice commands.

STOP after the Phase 16 final report.

WAIT FOR EXPLICIT INSTRUCTIONS FOR PHASE 17.